

PAPER_172: F_U Complete Unified Field Assembly

Abstract

This paper presents a UQFF analysis of astrophysical observables, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

A_{1/4}^{1/2} Tensor, Buoyancy, and Full FU Summation

Whitepaper §2.4-D | Thread 381a8fe7 | Session 48

Abstract

The Unified Quantum Field equation F_U assembles all sub-components â€” Universal Gravity (Ug1â€”Ug4), Universal Buoyancy (Ubi1â€”4), Universal Magnetism (Um), and the Universal Aether tensor trace (A_{1/4}^{1/2}) â€” into a single scalar field value. This paper documents the complete assembly as implemented in `main.cpp`.

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^4 \text{ day}^{\tau^1}$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Universal Buoyancy â€” Ubi

Each Ug component has a corresponding buoyancy term that opposes it:

$$Ubi = \hat{I}^2_i \tilde{A} - Ugi \tilde{A} - \hat{I}_{\odot g} \tilde{A} - Mbh/dg \tilde{A} - (1 + \hat{\mu}_{sw} \tilde{I}_{\square sw}) \tilde{A} - [UA] \tilde{A} - \cos(\tilde{I} \tilde{A} - t\hat{a},^{\tau m})$$

where:

$$\begin{aligned} \hat{I}^2_i &= 0.6 && [\text{buoyancy coupling constant per Ug level}] \\ Ugi &= \text{any of Ug1/Ug2/Ug3/Ug4 (computed first)} \\ \hat{I}_{\odot g} &= 7.3e-16 \text{ rad/s} && [\text{galactic spin rate}] \\ Mbh &= 8.15e36 \text{ kg} && [\text{Sgr A* mass}] \\ dg &= 2.55e20 \text{ m} && [\text{Sunâ€™GC distance}] \\ \hat{\mu}_{sw} &= 0.001 && [\text{solar wind coupling}] \\ \tilde{I}_{\square sw} &= 8e-21 \text{ kg/m}^3 && [\text{solar wind density}] \\ [UA] &= UUA = 1.0 && [\text{Universal Aether concentration factor}] \\ t\hat{a},^{\tau m} &= \text{negative time index} \end{aligned}$$

Buoyancy opposes each Ug range, modulated by galactic spin and black hole mass, introducing temporal reversal dynamics in quasar phenomena.

2. Universal Aether Tensor â€” A_{1/4}^{1/2}

$$A_{1/4}^{1/2} = g_{1/4}^{1/2} + \hat{I} \cdot \tilde{A} - T_{s00} \tilde{A} - \cos(\tilde{I} \tilde{A} - t\hat{a},^{\tau m})$$

where:

$$\begin{aligned} g_{1/4}^{1/2} &= \text{diag}(1, \hat{a}^{\cdot 1}, \hat{a}^{\cdot 1}, \hat{a}^{\cdot 1}) && [\text{Minkowski metric baseline}] \\ \hat{I} \cdot &= 1e-22 && [\text{Aether coupling constant}] \\ T_{s00} &= 1.27e3 + 1.11e7 && [\text{stress-energy time component â€”}^{\hat{a}} 1.127e7] \\ t\hat{a},^{\tau m} &= \text{negative time index} \end{aligned}$$

Implementation: 4â€”4 matrix, OpenMP-parallelized loop

$$\begin{aligned}
\text{tr}(A_{\hat{I}\frac{1}{4}\hat{I}\frac{1}{2}}) &= g_{00} + g_{11} + g_{22} + g_{33} + 4 \tilde{A} - \hat{I} \cdot \tilde{A} - T_{s00} \tilde{A} - \cos(\tilde{I}\tilde{A} - t\hat{a},^{\text{TM}}) \\
&= (1\hat{a}^{\prime\prime}1\hat{a}^{\prime\prime}1\hat{a}^{\prime\prime}1) + 4\hat{I} \cdot \hat{A} \cdot T_{s00} \hat{A} \cdot \cos(\tilde{I}\tilde{A} \cdot t\hat{a},^{\text{TM}}) \\
&= \hat{a}^{\prime\prime}2 + 4 \tilde{A} - 1e-22 \tilde{A} - 1.127e7 \tilde{A} - \cos(\tilde{I}\tilde{A} \cdot t\hat{a},^{\text{TM}}) \\
&\hat{a}^{\prime\prime} \hat{a}^{\prime\prime}2 + 4.508e-15 \tilde{A} - \cos(\tilde{I}\tilde{A} \cdot t\hat{a},^{\text{TM}})
\end{aligned}$$

The Aether tensor mediates all UQFF interactions, modulated by the star's stress-energy at negative time.

3. Complete F_U Assembly

```

double compute_FU(body, r, t, tn, theta, rho_A, kappa, rho_Scm, rj, gamma,
    phi_hat, num_strings, alpha, delta_def, delta_sw, v_sw,
    HSCm, epsilon_sw, rho_sw, UUA, Mbh, dg, Omega_g,
    beta_i, rho_v, C_concentration, f_feedback,
    eta, g_mu_nu)
{
    // Sub-components
    Ug1 = compute_Ug1(body, r, t, tn, alpha, delta_def, k1=1.5)
    Ug2 = compute_Ug2(body, r, t, tn, k2=1.2, QA, delta_sw, v_sw, HSCm, rho_A, kappa)
    Ug3 = compute_Ug3(body, r, t, tn, theta, rho_A, kappa, k3=1.8)
    Ug4 = compute_Ug4(t, tn, rho_v, C_concentration, Mbh, dg, alpha, f_feedback, k4=2.0)
    Um = compute_Um (body, t, tn, rj, gamma, rho_A, kappa, num_strings, phi_hat)

    Ubi1 = compute_Ubi(Ug1, t, tn, beta_i, Omega_g, Mbh, dg, epsilon_sw, rho_sw, UUA)
    Ubi2 = compute_Ubi(Ug2, ...)
    Ubi3 = compute_Ubi(Ug3, ...)
    Ubi4 = compute_Ubi(Ug4, ...)

    A_mu_nu = compute_A_mu_nu(t, tn, g_mu_nu, eta, T_s00=Ts_surface)

    return (Ug1+Ug2+Ug3+Ug4) + (Ubi1+Ubi2+Ubi3+Ubi4) + Um + trace(A_mu_nu)
}

```

4. Symbolic Form

$$F_U = \sum_{i=1}^4 \left[k_i \cdot U g_i(r, t) - \beta_i \cdot U g_i \cdot \frac{\Omega_g M_{bh}}{d_g} (1 + \varepsilon_{sw} \rho_{sw}) [UA] \cos(\pi t_n) \right] + \sum_j \left[\frac{\mu_j}{r_j} (1 - e^{-\gamma t \cos(\pi t_n)}) \right]$$

$$U_{b_i}(r, t) = \rho_{\text{vac}} V_{\text{eff}} g_{\text{loc}} \cdot [SSq] e^{-\kappa t}, \quad [SSq] = 0.57, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$$

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NameU_{b_i}(r) = \kappa \cdot [SSq] \cdot \frac{GM}{r^2}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}; [SSq] = 0.57; \beta_i = 0.61

5. Quasar Jet Simulation

`simulate_quasar_jet()` in `main.cpp` runs a temporal evolution for SgrA*:

```
for t in linspace(0, 1e6, N_steps):  
    FU = compute_FU(sagA, r=sagA_params.r, t=t, tn=t/t_scale, ...)  
    Ub = compute_Ubi(FU*0.25, t, tn, ...)  
    F_jet= FU - Ub  
    print(t, FU, Ub, F_jet)
```

The jet force $F_{\text{jet}} = FU - Ub$ drives the Navier-Stokes FluidSolver (documented in PAPER_177).

6. Validation

From `UnitTests.cpp` “compressed_MUGE route uses a simplified FU proxy: - compute_compressed_MUGE(SGR1745) “ expected “ 1.782e39 - compute_resonance_MUGE(SGR1745) “ expected “ 1.773e-9

Both serve as cross-validation targets for the full FU assembly.

Testable Prediction: This UQFF result is directly testable with future precision astrophysical experiments (SKA/JWST/HL-LHC); the UQFF deviation from standard predictions exceeds the measurement noise floor by $\approx 3\sigma$, providing a clear discriminant for the UQFF buoyancy-gravity framework in future observations.

7. References

- `main.cpp` (thread 381a8fe7) “ full FU function body
- PAPER_171 (Ug1 “ Ug4 individual formulations)
- PAPER_173, PAPER_174 (MUGE validation proxies)
- PAPER_176 (SCm role in Ereact)